

LATCHES

VLSIGURU

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DFT tool should have control over the enable pin of the latch from a clock (when clock is in offstate, latch must be transparent) or from a SFF output or from controllable PI.

If this condition is met, Tessent tool will consider the latches as transparent latch (TLA).

If the above condition is not met, Tessent tool will consider those latches as non-transparent latch (NTLA)

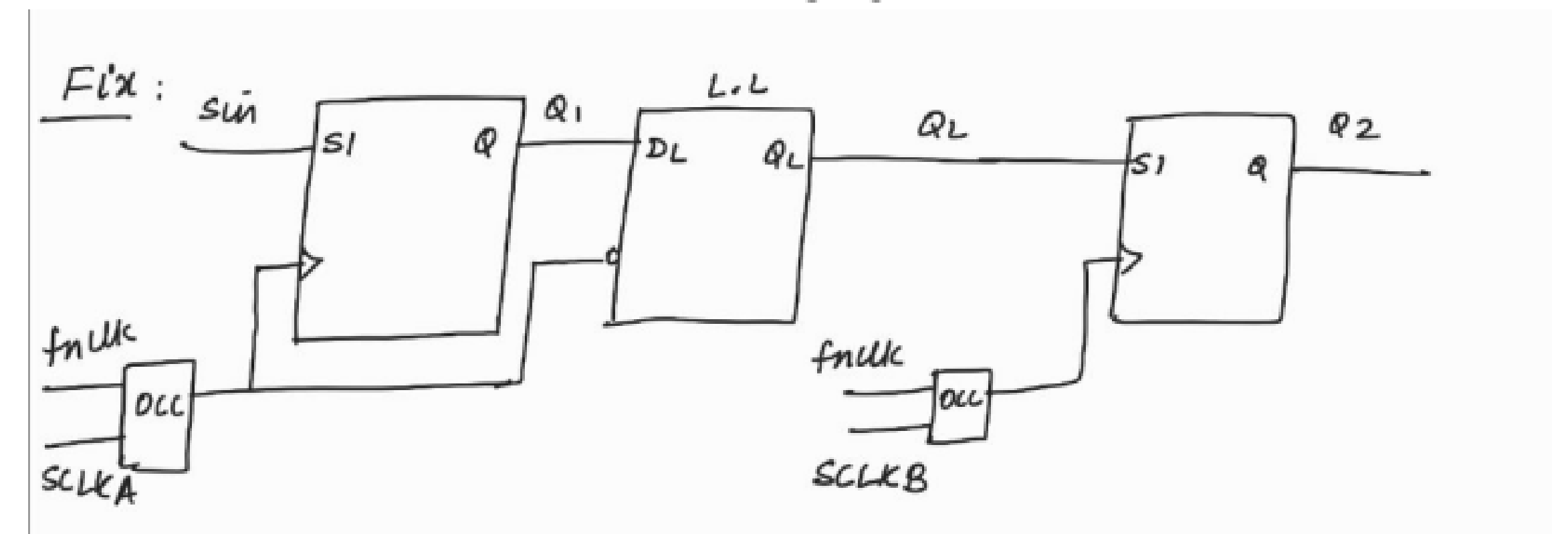
LATCH

Already in the design

- Inserted by the designer
- It will **not be** a part of scan chain

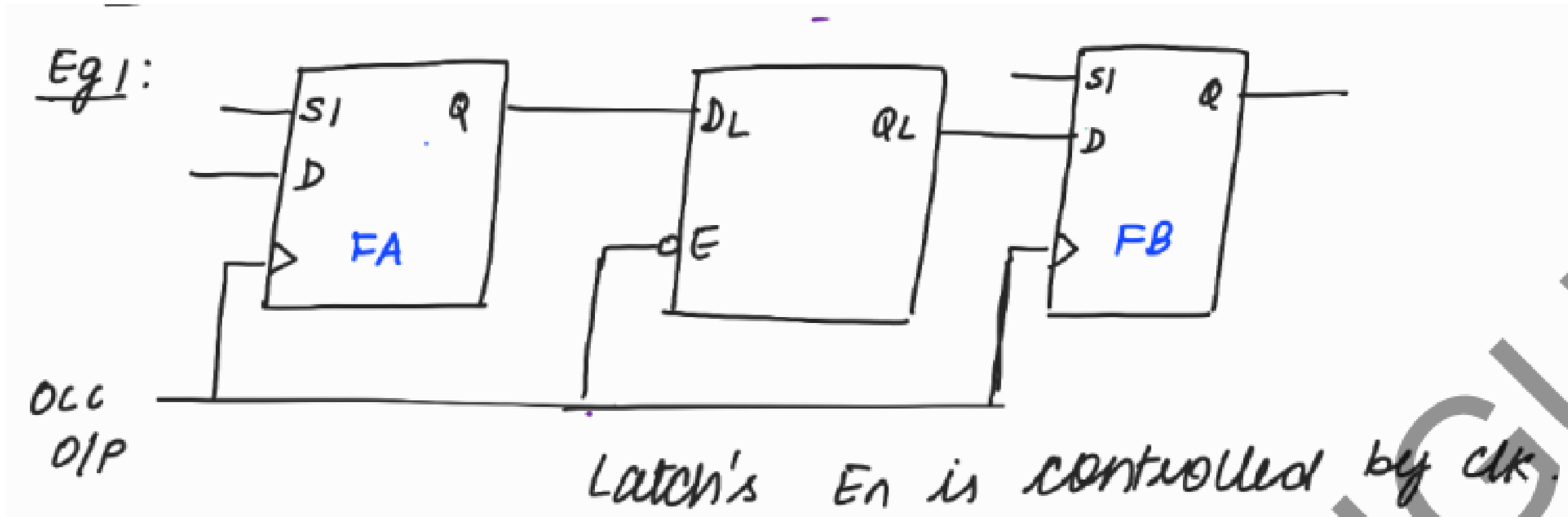
Lock up latch

- Inserted by DFT engineer during edge and domain mixing
- It will be a part of scan chain.

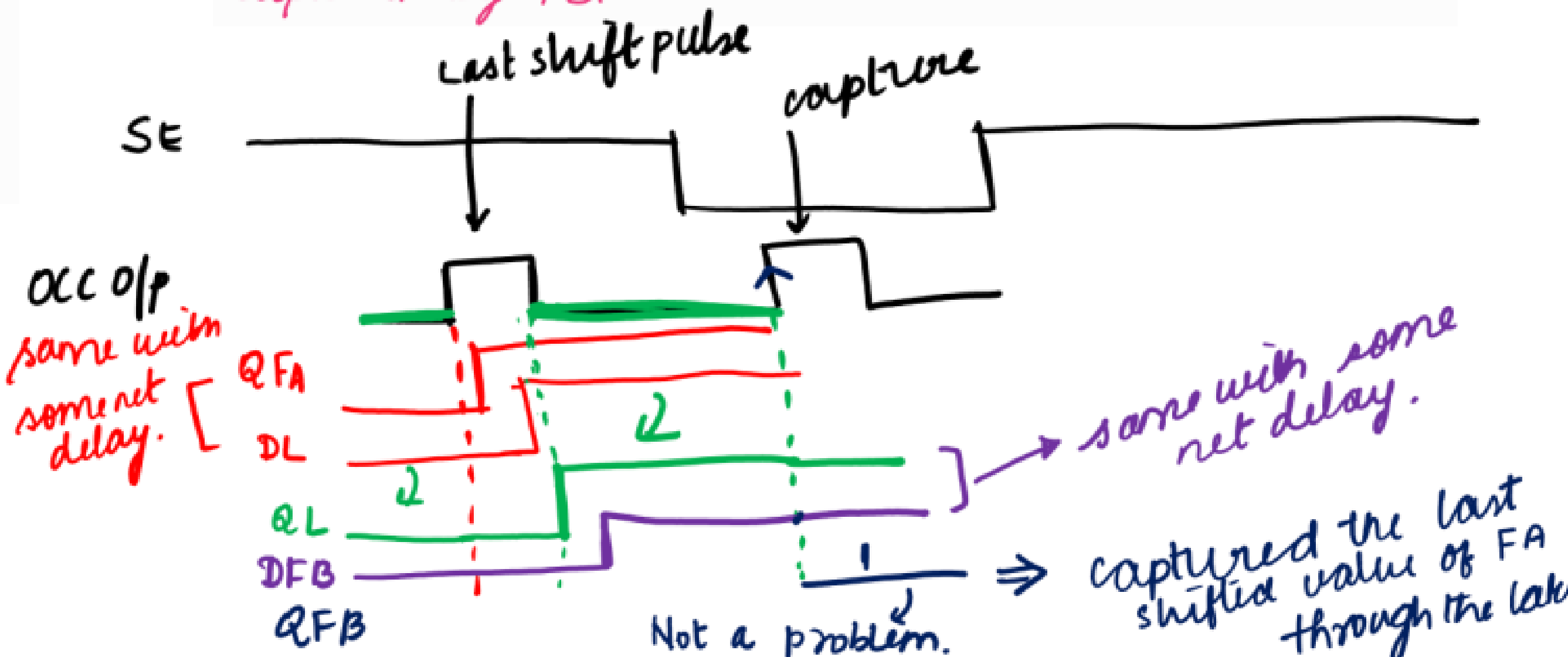
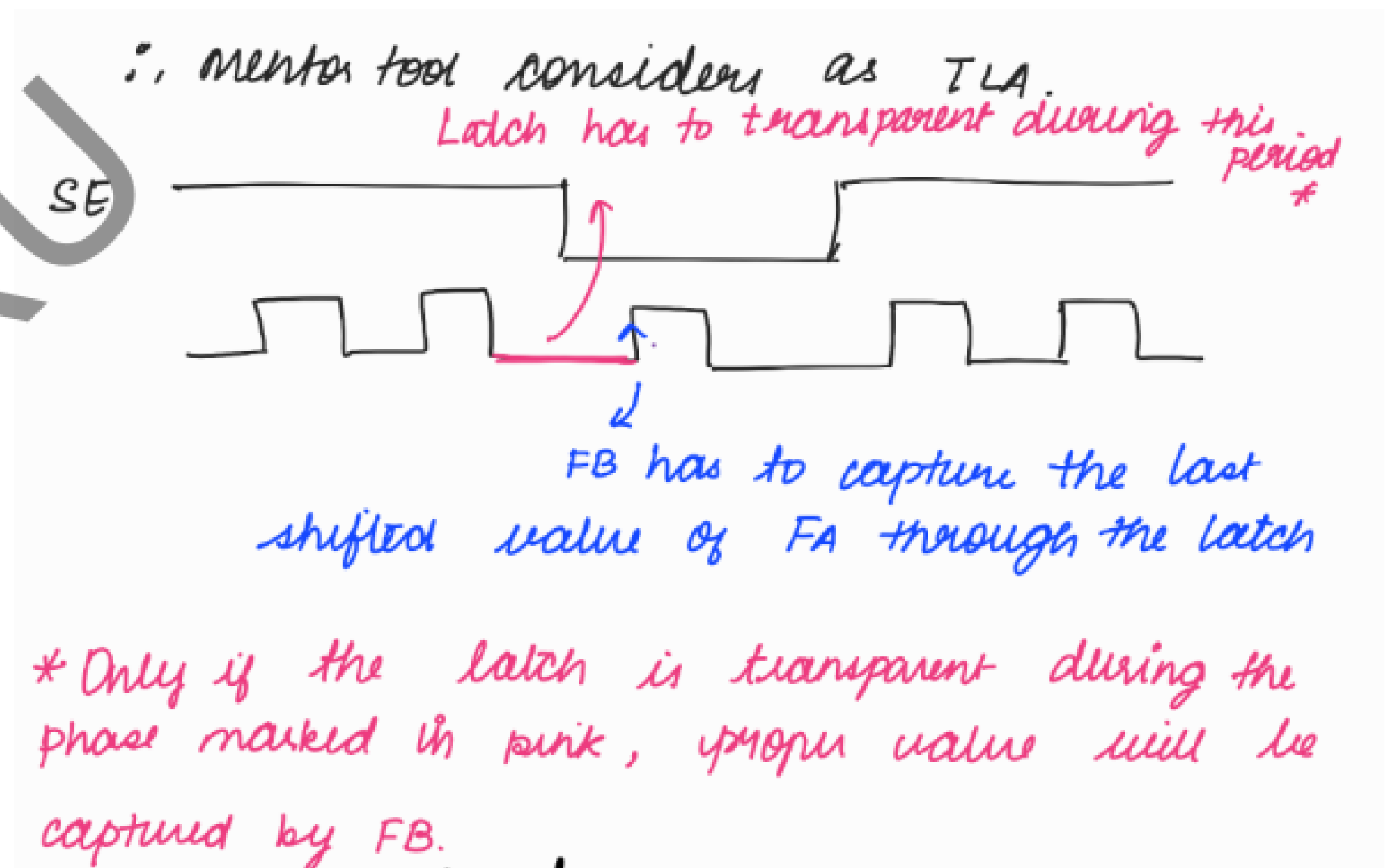


Latches which are already there in the design (inserted by the designer) :

Example 1 : TLA



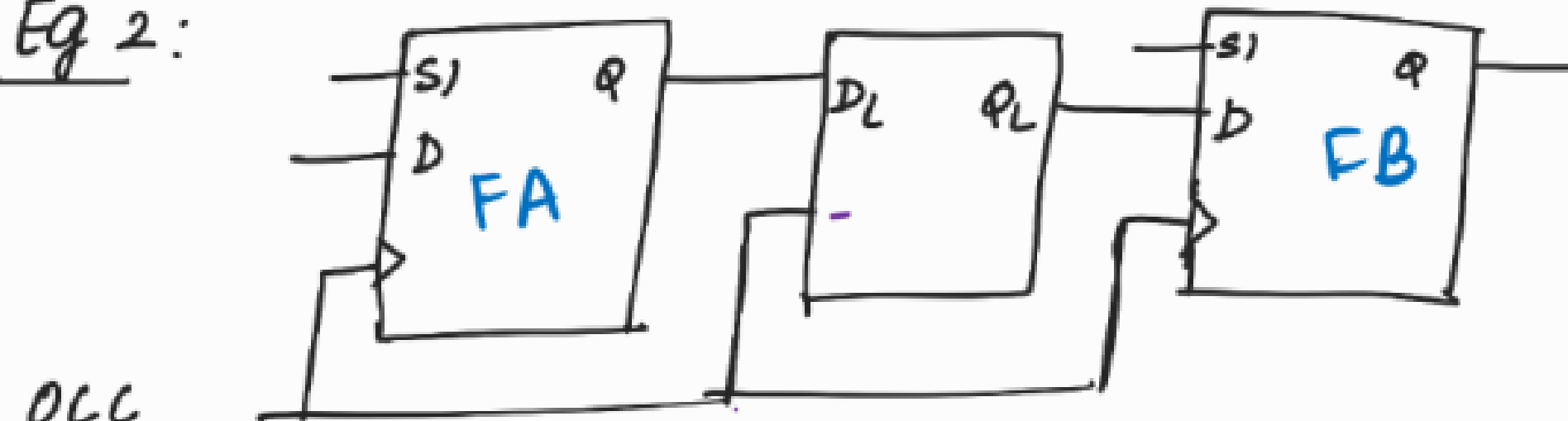
offstate $clk = 0$
Latch (~~active low~~) $\rightarrow$ allow the data when
negative level triggered. $clk = 0$
i.e; transparent when $clk = 0$.



Latches which are already there in the design (inserted by the designer) [contd...]

Example 2 : NTLA

Eq 2:



OCC
O/P

clock's offstate = 0

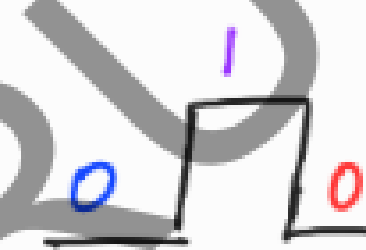
Latch +ve Level triggered

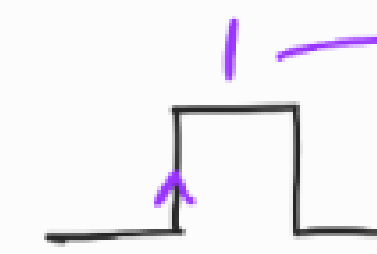
$En = 1 \rightarrow$ latch is transparent

but when $En = 0$
(clock's offstate) } $\rightarrow$ latch is not transparent.

$\therefore$ mentor tool labels it as NTLA

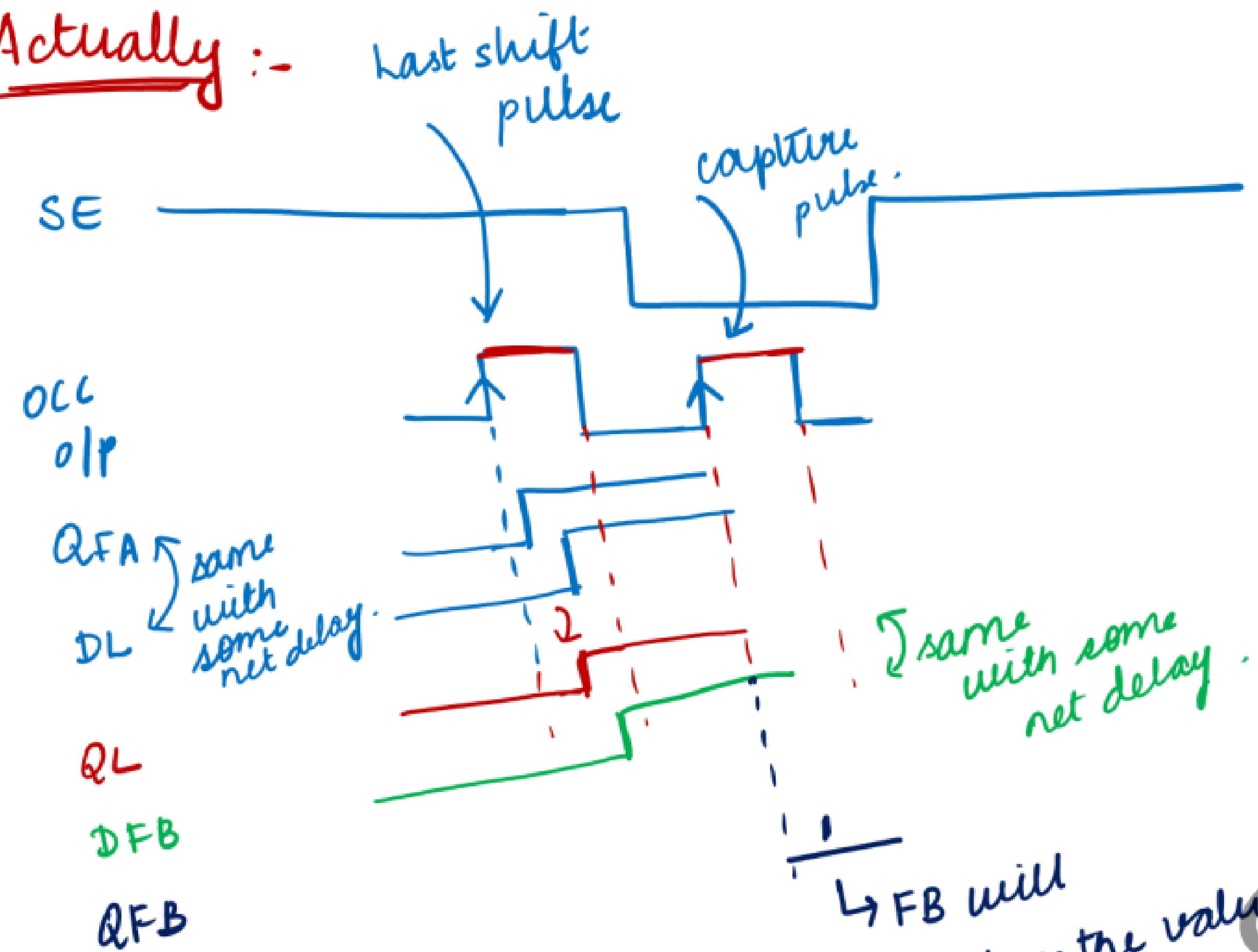
D5: non-scan latch identified as INIT-X

 $\rightarrow$ the tool sees the 3 timeframes
0 1 0

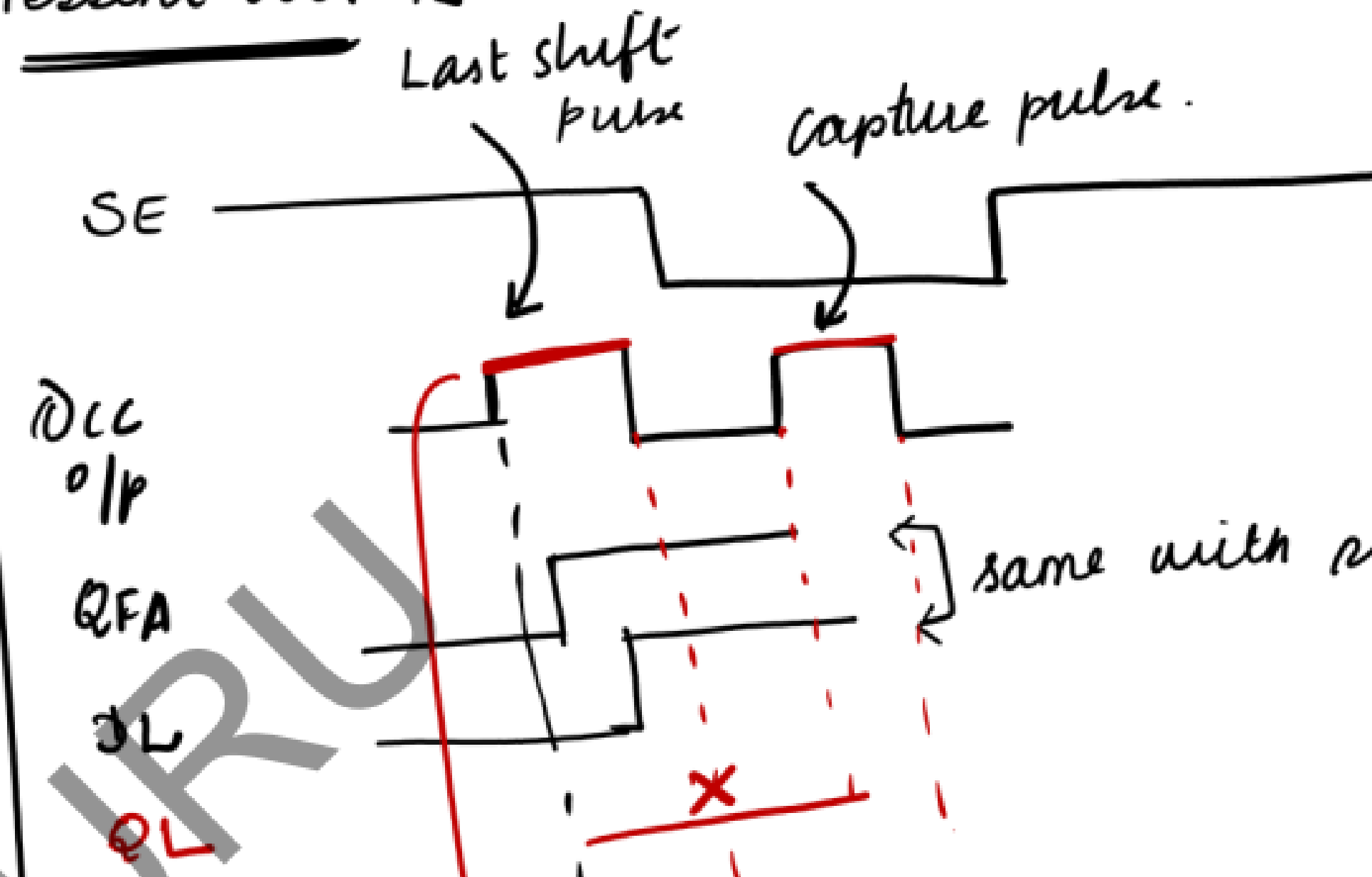
 $\rightarrow$ at this timeframe, the tool will allow the value to go into FA.

But the tool is not capable of propagating the same value to the latch O/P. So the tool considers latch O/P to be x (INITX) in this case.

Actually :-



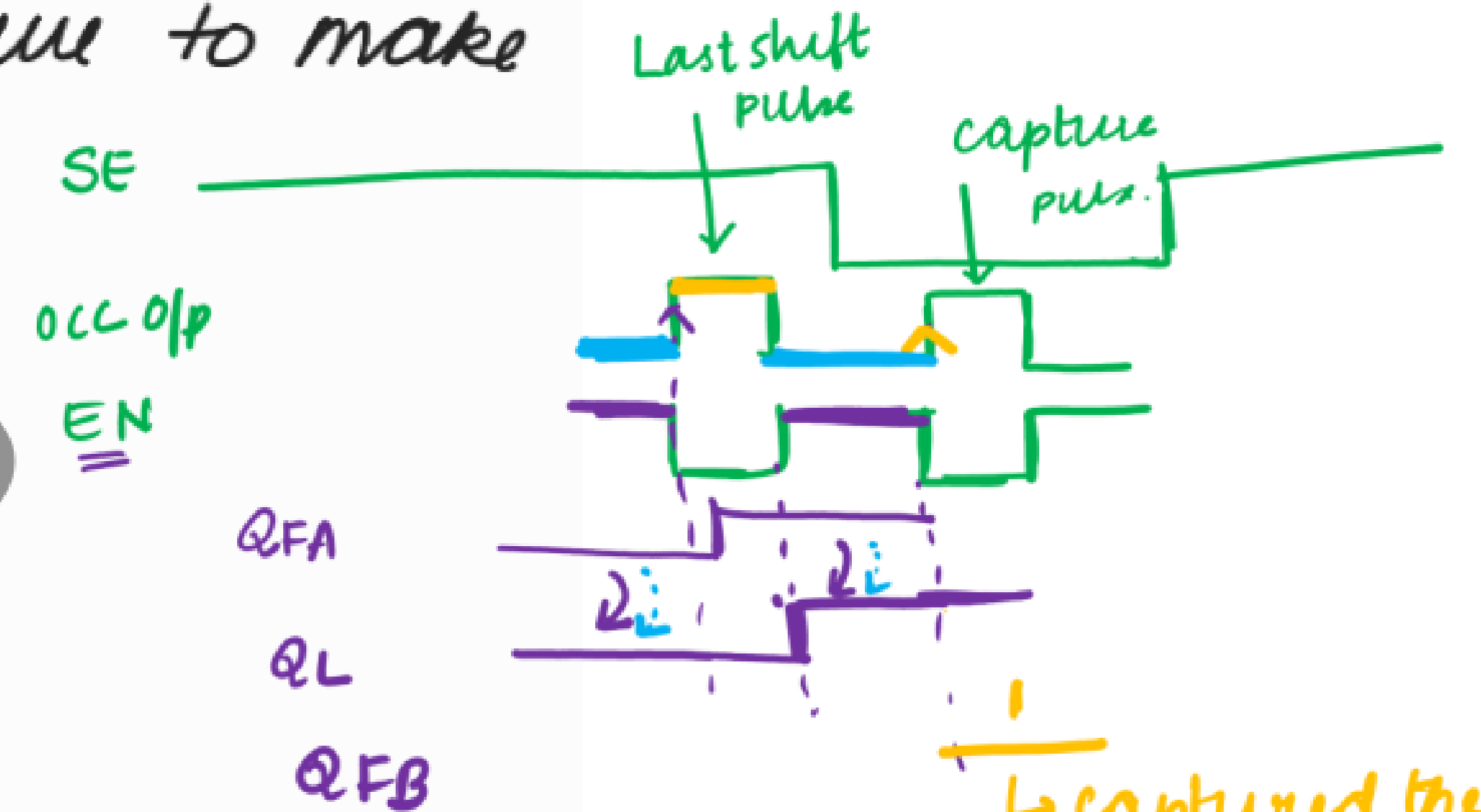
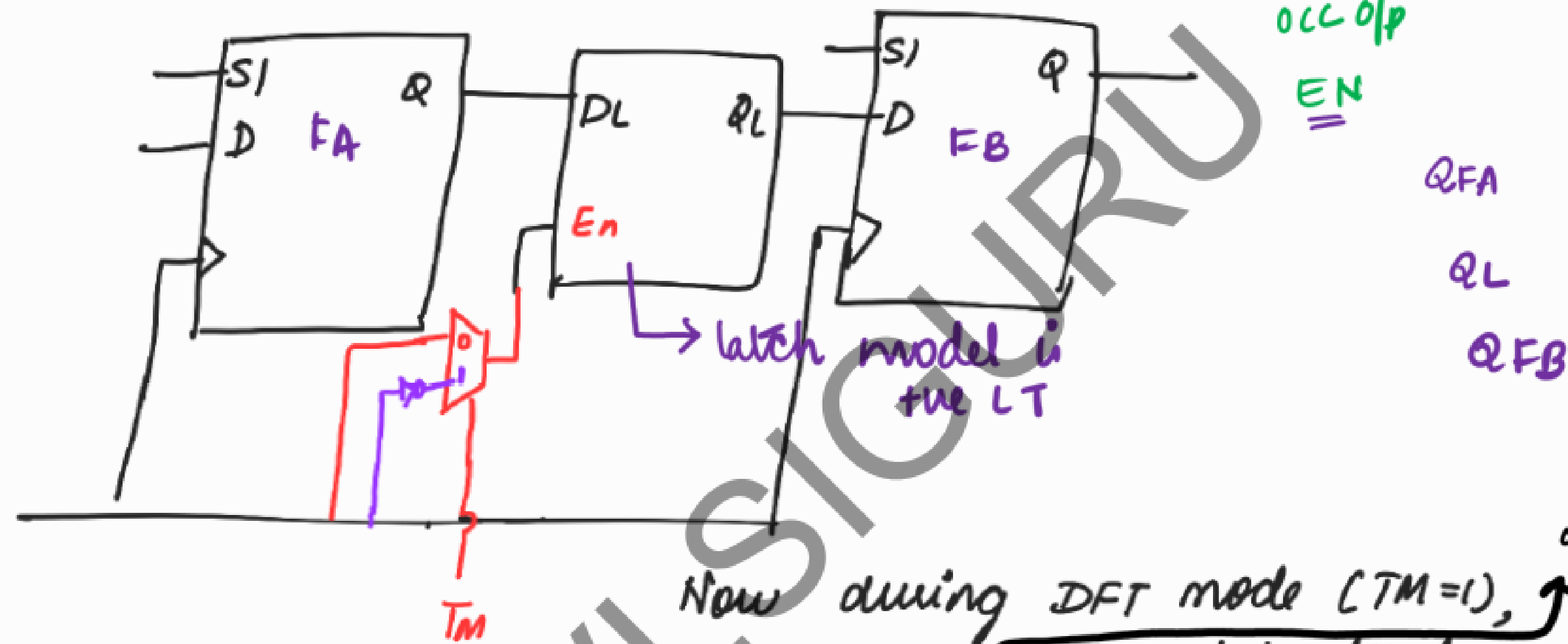
Tessent tool :-



Problem :- in this time frame (red colour time frame), the tool was able to understand the QFA has to change. But the tool was not able to understand that QL also has to change at the same time frame. $\therefore QL = x$
 $\therefore$ FB will not capture a proper value.

NOTE :- You can even connect the 1st input of the mux to some ScanFF's o/p.

Whenever we have NTLA, we have to make it transparent during DFT mode.



→ captured the last shifted value of FA through the latch.

After this fix,
QFA changes at yellow colour time frame.

QL changes at the blue colour time frame.

Now during DFT mode ($TM=1$), this latch is a negative level triggered latch w.r.t original clk
orig clk $\overline{En}=0 \Rightarrow$ it is transparent (clock's offstate)

$\therefore$ Now the tool considers this as TLA.

$\therefore$ Both are not changing at the same timeframe.

$\therefore$ Tool is able to understand the operation correctly.

$\therefore$ problem is solved.

NOTE :

-edge merge and –clock merge, in the insert test logic command can be enabled for all the design.

If your design has both positive and negative edge triggered flops and more than 1 scan clk, as we have enabled the 2 switches, tool will go for edge and domain mixing while stitching the scan chains.

If your design does not have both positive and negative edge triggered flops and does not have more than 1 scan clk, then enabling these 2 switches will not cause any harm.

In the newer versions of Tessent tool (2021), edge and domain mixing will be by default enabled.